Assignment #3

Due Date: March 6-th 2026

Objectives

- Dynamic Programming

Submission Instructions

- Write your answer to problem 1 and part a of problem 2 in a file named $A3.pdf$.
- Write your answer to part b of problem 2 in a file name $P2.py$.
- Submit both $A3.pdf$ and $P2.py$ to the corresponding dropbox folder. Do not create any folder
- Make sure your writing is neat or you type.
- Make sure to include the following at the beginning of every file you submit.

I certify that this assignment is entirely my own work and that all external resources, if used, have been properly acknowledged.

Problem 1 [10 points]

We have several coins arranged in a row next to each other. The i-th coin has value c_i. Two persons take coins from this row in the following way:

- The first person picks one coin
- The second person must pick the coin immediately before and after the coin taken by the first one.

The first person has a constraint. The total value of the coins they pick must not exceed k. what is the maximum total value that the first person can obtain? Develope the recurrence relation, provide the pseudocode (memoized version), show a few rows and columns of the DP table, and analyze the time complexity of your algorithm.

Problem 2 [15 points]

A company offers a product in packages of different sizes. You are given a list of the available package sizes. You want to purchase exactly **m** units of the product. Design a dynamic programming algorithm to determine:

- The minimum number of packages required, and
- The sizes of the packages you must purchase

so that the total number of units is exactly **m**.

If it is not possible to obtain exactly m units using the given package sizes, your algorithm should return –1.

(a) Write the recurrence relation and provide a small DP table trace for the input in the first example given below.

(b) Write the code of the program to solve this problem. The program file should be named *P2.py*, read its input from a file whose name is provided as a command-line argument, and print the output to standard output (the screen).

Ensure that both input and output exactly follow the format below:

- Input Format: The content of input.txt

```
Line 1: A comma-separated list of package  sizes.
Line 2: A single integer m, the target amount.
```

- Output Format:

```
Line 1: The total number of packages.
Line 2: The quantity of packages for each size.
```

Here are how to run the program and a few example of inputs and outputs

```
>>python P2.py input.txt
content of input.txt:
2,3
6

output:
2
2 packages of size 3
|-----------------------------------------------------------
content of input.txt
6,9,20
8

output:
-1
There's no solution
|-----------------------------------------------------------
```

```
content of input.txt
1,5,12
15

output:
3
3 packages of size 5
|-----------------------------------------------------------
content of input.txt
1,3,4
7

output:
2
1 packages of size 3 and 1 package of size 4
```